AyuAware Exposure Section

Technical Product Requirements Document

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1. Executive Summary

What We're Building

The Exposure section tracks **external environmental factors** that influence health according to Ayurvedic principles. It answers the question: "*What's in my environment right now, and how is it affecting my body?*"

This section integrates three categories of external influences:

Category	Data Sources	Ayurvedic Relevance
Environmental	Weather, temperature, humidity, air quality	Directly aggravates/pacifies doshas
Light	UV index, lux levels, sunrise/sunset	Affects circadian rhythm, Ojas, Vitamin D
Natural Cycles	Lunar phases, (optional) planetary transits	Influences energy, mood, bodily rhythms

Why It Matters

In Ayurveda, the external environment is not separate from internal health — it directly shapes it:

- **Hot, humid weather** aggravates Pitta (inflammation, irritability)
- **Cold, dry conditions** aggravate Vata (anxiety, dryness)

- **Cold, wet weather** aggravates Kapha (congestion, sluggishness)
- **Lunar cycles** influence energy, digestion, and mental state
- **Light exposure** governs circadian rhythm and vitality

Most people don't connect their symptoms to environmental factors. AyuAware makes this connection explicit and actionable.

Key Differentiator

Weather apps show conditions. AyuAware shows **personal impact**:

- Not just "28°C and humid" but "This weather may aggravate your Pitta, causing irritability and inflammation"
- Not just "UV index 7" but "Good time for 15 minutes of sun exposure for Vitamin D"
- Not just "Waning moon" but "Lower energy is natural — focus on rest and completing projects"

Every data point connects to the user's specific constitution and provides actionable guidance.

2. Section Overview

2.1 Information Architecture

```
Exposure Section
│
├── 🌤️ Today's Environmental Impact
│    ├── Current weather conditions
│    ├── Temperature & humidity
│    ├── Air quality (AQI)
│    ├── Personal dosha impact
│    └── Recommendations
│
├── ☀️ Light Exposure
│    ├── UV index (current & forecast)
│    ├── Lux levels
│    ├── Sunrise/sunset times
│    ├── Circadian recommendations
│    └── Vitamin D guidance
│
├── 🎯 Today's Focus
│    └── Actionable checklist based on conditions
│
├── 🌙 Lunar Influence
│    ├── Current moon phase
│    ├── Phase meaning & effects
│    ├── Dosha impacts
│    └── Recommendations
```

```
│
└── 🪐 Planetary Influence (Optional)
    ├── Current transits
    ├── Vedic astrology perspective
    ├── Timing guidance
    └── Recommendations
```

2.2 The Exposure Philosophy

Environment as Medicine (or Poison)

Ayurveda teaches that the same environment affects different constitutions differently:

Constitution	Hot Weather	Cold Weather	Humid Weather
Vata	Pacifying ✓	Aggravating ⚠️	Neutral
Pitta	Aggravating ⚠️	Pacifying ✓	Aggravating ⚠️
Kapha	Pacifying ✓	Aggravating ⚠️	Aggravating ⚠️

The Twin interprets conditions through the lens of the user's specific constitution.

Cycles Within Cycles

The section tracks multiple overlapping cycles:

- **Daily**: Sunrise/sunset, UV patterns, temperature curves
- **Monthly**: Lunar phases (29.5-day cycle)
- **Seasonal**: Weather patterns, daylight hours
- **Optional**: Planetary transits (for those interested in Vedic astrology)

2.3 Design Philosophy

Personalized Impact First Don't just show data — show what it means for *this user*. The impact alert is the first thing they see.

Actionable Checklist Today's Focus distills all environmental factors into a simple checklist of priority actions.

Progressive Disclosure

- Essential: Environmental impact, Today's Focus
- Detailed: Light exposure, lunar influence
- Optional: Planetary influences (can be disabled)

Educational Integration First-time visitors learn why environment matters in Ayurveda. Concepts are introduced naturally.

Protocol & Shop Connection Every insight can lead to a protocol or relevant products (cooling supplements, Vitamin D, etc.).

3. User Journey Breakdown

3.1 First-Time Visitor Experience

Context: User navigates to Exposure section for the first time.

Twin Message 1:

"Your environment plays a huge role in your health. Let me show you what's affecting you today."

Educational Card:

```
┌──────────────────────────────────────────────┐
│ 💡 What is Exposure?                          │
│ ──────────────────────────────────────────────│
│                                               │
│ In Ayurveda, your external environment        │
│ directly influences your internal state.      │
│                                               │
│ I track three key exposures:                  │
│                                               │
│ 🌤️ Environment                                │
│ Weather, air quality, light levels            │
│                                               │
│ 🌙 Lunar Cycles                               │
│ Moon phases and their health impacts          │
│                                               │
│ 🪐 Planetary Influences (Optional)            │
│ Astrological transits and timing              │
│                                               │
│ These factors affect your doshas and can      │
│ aggravate or pacify imbalances.               │
│                                               │
│                [Got it! →]                    │
└──────────────────────────────────────────────┘
```

After user taps "Got it!", section loads with full data.

3.2 Returning User Experience

Context: User returns to Exposure section.

Twin Message:

"Let's see what environmental factors are affecting you today..."

Immediately shows Today's Environmental Impact card.

3.3 Exploring Environmental Impact

Context: User views the main Environmental Impact card.

Collapsed Environmental Impact Card:

```
┌──────────────────────────────────────────────────┐
│ 🌤️ Today's Environmental Impact                  │
│ ──────────────────────────────────────────────── │
│                                                  │
│        [Weather Icon: Partly Cloudy]             │
│     Warm, Humid, Partly Cloudy                   │
│                                                  │
│ Impact on You:                                   │
│  ┌──────────────────────────────────────────┐    │
│  │ ⚠️ This weather may aggravate Pitta       │    │
│  │                                          │    │
│  │ Why: High heat and humidity increase     │    │
│  │ internal heat, possibly causing irritability │
│  │ and inflammation.                        │    │
│  └──────────────────────────────────────────┘    │
│                                                  │
│ Key Metrics:                                     │
│ 🌡️ 28°C  •  💧 81%  •  🌬️ AQI 44 (Good)          │
│                                                  │
│          [See Full Details ↓]                    │
└──────────────────────────────────────────────────┘
```

Key Elements:

- **Weather visualization** — Animated icon showing current conditions
- **Impact alert** — Color-coded by affected dosha (coral for Pitta)
- **Quick metrics** — Temperature, humidity, AQI at a glance

User Taps "See Full Details":

Expanded Environmental Impact Card:

1. **Location & Update Time**
    - "Your Location: Dharamsala, India"
    - "Last Updated: 5 minutes ago"
2. **Impact on Your Constitution**
    - Dosha being affected (Pitta in this example)
    - Why this happens (heat + humidity = internal heat)
    - Specific symptoms to watch for:

        - Irritability and short temper
        - Skin inflammation or rashes
        - Increased appetite and thirst
        - Digestive upset (acid reflux)
        - Disturbed sleep quality
3. **Detailed Conditions**
    - Temperature: 28°C (feels like 32°C)
    - Humidity: 81% (high, Kapha-aggravating)
    - Air Quality: AQI 44 (Good, safe outdoors)
    - Weather Pattern: 20% rain chance, wind 8 km/h
    - Pressure: 1013 hPa (stable)
4. **Recommendations for Today**
    - Stay in shade during peak sun (12-3pm)
    - Drink cooling beverages (coconut water)
    - Wear light, breathable clothing
    - Avoid spicy, heating foods
    - Practice cooling pranayama (Sheetali)
    - Apply cooling oils (coconut, sunflower)
5. **Action Buttons**
    - [Create Weather Protocol →]
    - [View 7-Day Forecast →]
    - [Shop Cooling Products →]

3.4 Exploring Light Exposure

Context: User scrolls to Light Exposure card.

Twin Message:

"Light exposure affects your circadian rhythm and Ojas. Here's today's light environment:"

Collapsed Light Exposure Card:

```
┌──────────────────────────────────────────────┐
│ ☀️ Light Exposure                            │
│ ──────────────────────────────────────────── │
│                                              │
│ [Sun Arc Visualization showing current position]
│                                              │
│ 🌅 Sunrise: 6:41 AM                          │
│ 🌇 Sunset: 7:42 PM                           │
│                                              │
│ Current UV Index: Low (2)                    │
│ Lux Level: 1,000 lux                         │
│                                              │
│ 💡 Insight: Good time for outdoor exposure   │
```

```
without sun protection

            [See Full Details ↓]
```

Expanded Light Exposure Card:

1. **Daylight Information**
    - Total daylight: 13h 1m
    - Brahma Muhurta: 5:11 AM - 6:41 AM (best for meditation)
    - Twilight: 7:42 PM - 8:12 PM
2. **UV Index Throughout Day**
    - Line chart showing UV curve
    - Current: 2 (Low)
    - Peak: 7 (High) at 1:00 PM
    - Protection needed: 11 AM - 3 PM
3. **Light Intensity (Lux)**
    - Current: 1,000 lux
    - Reference scale:
        - Direct sunlight: 32,000-100,000 lux
        - Indirect daylight: 10,000-25,000 lux
        - Office lighting: 320-500 lux
        - Home lighting: 50-150 lux
4. **Impact on Health**
    - Circadian rhythm recommendations
    - Vitamin D production guidance
    - Ojas (vitality) impact
5. **Recommendations**
    - Get 15 min morning sun (before 9 AM)
    - Work near windows during day
    - Use sunscreen if outdoors 11 AM - 3 PM
    - Dim lights 2 hours before bed
    - Avoid screens after sunset (use filters)
6. **Action Buttons**
    - [Create Circadian Protocol →]
    - [Shop Light Therapy →]
    - [Shop Vitamin D →]

3.5 Using Today's Focus Checklist

Context: User views the actionable checklist.

Twin Message:

"Based on today's conditions, here are the most important things to focus on:"

Today's Focus Card:

```
🎯 Today's Focus
──────────────────────────────────────────

Priority Actions for Your Health:

☐ Take Vitamin D After Lunch
  UV is low, supplement recommended

☐ Stay Hydrated (3 Glasses of Water)
  High humidity increases Kapha

☐ Eat Main Meal at Noon
  Optimal digestion window

☐ Gentle Evening Activities Only
  Waning moon reduces vitality

Progress: 0/4 completed

          [Mark All Complete]
```

Interaction:

- Tap checkbox to mark complete
- Checkbox fills with animation (200ms)
- Text gets strikethrough
- Progress updates ("1/4 completed")
- Completion tracked for the day

Source of Items:

- Items are generated based on:
  - Weather conditions
  - UV levels
  - Moon phase
  - User's constitution
  - Current imbalances (from Vikriti)
  - Active protocols

3.6 Understanding Lunar Influence

Context: User scrolls to Lunar Influence card.

Twin Message:

"The moon's cycles influence your body's rhythms. Let's see what phase we're in:"

Collapsed Lunar Card:

```
🌙 Lunar Influence
──────────────────────────────────────────

        [Moon Phase Visualization]
         Waning Crescent Moon

Current Phase:
Waning Crescent Moon

Impact: May reduce vitality

This phase can lower natural energy and Ojas.
Focus on rest and gentle activities.

           [See Full Details ↓]
```

Expanded Lunar Card:

1. **Phase Information**
    - Current: Waning Crescent
    - Day 26 of 29 in lunar cycle
    - Next New Moon: In 3 days (Feb 3)
2. **What This Phase Means**
    - Educational explanation of waning crescent
    - Time of rest and renewal before new cycle
    - Energy naturally decreases as moon wanes
3. **Impact on Doshas**
    - Vata: Stable (waning calms excess movement)
    - Pitta: Reduced (favorable for Pitta types)
    - Kapha: Slightly elevated (lower energy increases heaviness)
4. **Health Effects**
    - Lower natural energy (not pathological)
    - Desire for introspection and rest
    - Reduced appetite
    - Better for finishing vs. starting projects
    - Deeper sleep
5. **Ayurvedic Perspective**
    - Moon governs Kapha and water element
    - Waning moon reduces Bala (strength) and Ojas
    - Nature's way of encouraging rest
6. **Full Lunar Cycle Visualization**
    - 8 phase icons showing complete cycle
    - Current phase highlighted

    - Dates for upcoming phases
7. **Recommendations**
    - Prioritize rest and sleep (9+ hours)
    - Gentle yoga and walking
    - Complete ongoing projects
    - Practice meditation and reflection
    - Lighter meals
    - Limit social commitments
8. **Supplements to Support**
    - Ashwagandha for rest and recovery
    - Magnesium for deep sleep
    - Brahmi for calm introspection
9. **Action Buttons**
    - [Create Moon Phase Protocol →]
    - [View Full Lunar Calendar →]
    - [Shop Moon Cycle Support →]

3.7 Planetary Influence (Optional)

Context: User scrolls to Planetary Influence card (if enabled).

Twin Message:

"For those interested in Vedic astrology, planetary positions also influence health:"

Collapsed Planetary Card:

```
┌──────────────────────────────────────────────────┐
│ 🪐 Planetary Influence (Optional)                │
│ ────────────────────────────────────────────────│
│                                                  │
│        [Astrological Symbol]                     │
│                                                  │
│ Current Transits:                                │
│ ☉ Sun in Gemini                                  │
│ ♄ Saturn in Transit                              │
│                                                  │
│ Impact: Mercury rules Gemini, affecting          │
│ communication and mind. Saturn's transit may     │
│ bring lessons.                                   │
│                                                  │
│           [See Full Details ↓]                   │
│           [Disable Planetary Info]               │
└──────────────────────────────────────────────────┘
```

Important Note:

- This section is **optional**
- Users can disable via "Disable Planetary Info" button
- Preference is persisted
- Some users prefer purely Ayurvedic approach without astrology

Expanded Planetary Card:

1. **Vedic Astrology Perspective**
    - Brief explanation of Jyotish
    - How planetary positions influence health
2. **Current Major Transits**
    - Sun in Gemini (Mercury-ruled)
        - Mental activity increases
        - Communication important
        - Restlessness may arise (Vata)
    - Saturn transit
        - Lessons through discipline
        - Joint/bone issues may surface
        - Slow but steady progress
3. **Timing Considerations**
    - Favorable times today (for meditation, meals)
    - Unfavorable times (Rahu Kala)
4. **Action Buttons**
    - [View Your Birth Chart →]
    - [Create Astrological Protocol →]
    - [Disable Planetary Info]

3.8 Creating an Environmental Protocol

Context: User taps "Create Weather Protocol" from Environmental Impact card.

Twin Message:

"Let me create a protocol to help you balance today's warm, humid weather..."

Protocol Card:

```
┌──────────────────────────────────────────────┐
│ 🌤️ Pitta-Balancing Weather Protocol          │
│ ──────────────────────────────────────────── │
│                                              │
│ 🍽️ Nutrition                                 │
│ ──────────────────────────────────────────── │
│ • Favor cooling foods (cucumber, mint)       │
│ • Drink coconut water throughout day         │
│ • Avoid spicy, fried, fermented foods        │
│ • Eat at regular times (not when heated)     │
```

```
│                                                                │
│  [View Cooling Meal Ideas →]                                   │
│                                                                │
│  💊 Supplements                                                │
│  ──────────────────────────────────────────────────────────    │
│  • Aloe Vera juice (cooling)                                   │
│  • Brahmi (calms Pitta mind)                                   │
│  • Coconut oil capsules                                        │
│                                                                │
│  [Shop Cooling Supplements →]                                  │
│                                                                │
│  🧘 Lifestyle                                                  │
│  ──────────────────────────────────────────────────────────    │
│  • Stay in shade 12-3 PM                                       │
│  • Cool shower in evening                                      │
│  • Practice Sheetali pranayama (cooling)                       │
│  • Wear light, loose clothing                                  │
│  • Coconut oil massage before bed                              │
│                                                                │
│  Duration: Use while hot weather lasts                         │
│  Difficulty: Easy                                              │
│                                                                │
│  [Save to My Protocols]  [Start Now →]                         │
└────────────────────────────────────────────────────────────────┘
```

Protocol Types:

- Weather-based (responds to current conditions)
- Lunar-based (aligned with moon phases)
- Circadian (light/sleep focused)
- Planetary (if astrology enabled)

4. Technical Architecture

4.1 High-Level Architecture

```
┌──────────────────────────────────────────────────────────────────
```

```
│   │   ┌──────────────┐  ┌──────────────┐  ┌──────────────┐
┌──────────────┐   │   │
│   │   │ Environ-     │ │   Light      │ │  Today's     │ │   Lunar/
│   │   │              │
│   │   │ mental       │ │  Exposure    │ │   Focus      │ │  Planetary
│   │   │              │
│   │   └──────────────┘  └──────────────┘  └──────────────┘
└──────────────┘   │   │
│
└──────────────────────────────────────────────────────────────────────────────
─┘   │
└──────────────────────────────────────────────────────────────────────────────
──────────┘
                                          │
                                          ▼
┌──────────────────────────────────────────────────────────────────────────────
──────────┐
│                                STATE LAYER
│
│   ┌────────────────────┐   ┌────────────────────┐
┌────────────────────┐
│   │  Weather Store     │   │  Cycles Store      │   │  Exposure UI
│   │                    │   │                    │   │  Store
│   │• Conditions        │   │• Lunar phase       │   │• Expanded
cards │   │
│   │• Temperature       │   │• Planetary         │   │• Focus
items     │   │
│   │• Humidity          │   │• Sunrise/sunset    │   │•
Preferences     │   │
│   │• AQI               │   │• UV forecast       │   │• Messages
│   │                    │   │                    │
│   └────────────────────┘   └────────────────────┘
└────────────────────┘   │
└──────────────────────────────────────────────────────────────────────────────
──────────┘
                                          │
                                          ▼
┌──────────────────────────────────────────────────────────────────────────────
──────────┐
│                               SERVICE LAYER
│
│   ┌──────────────────────┐
┌──────────────────────────────────────┐     │
```

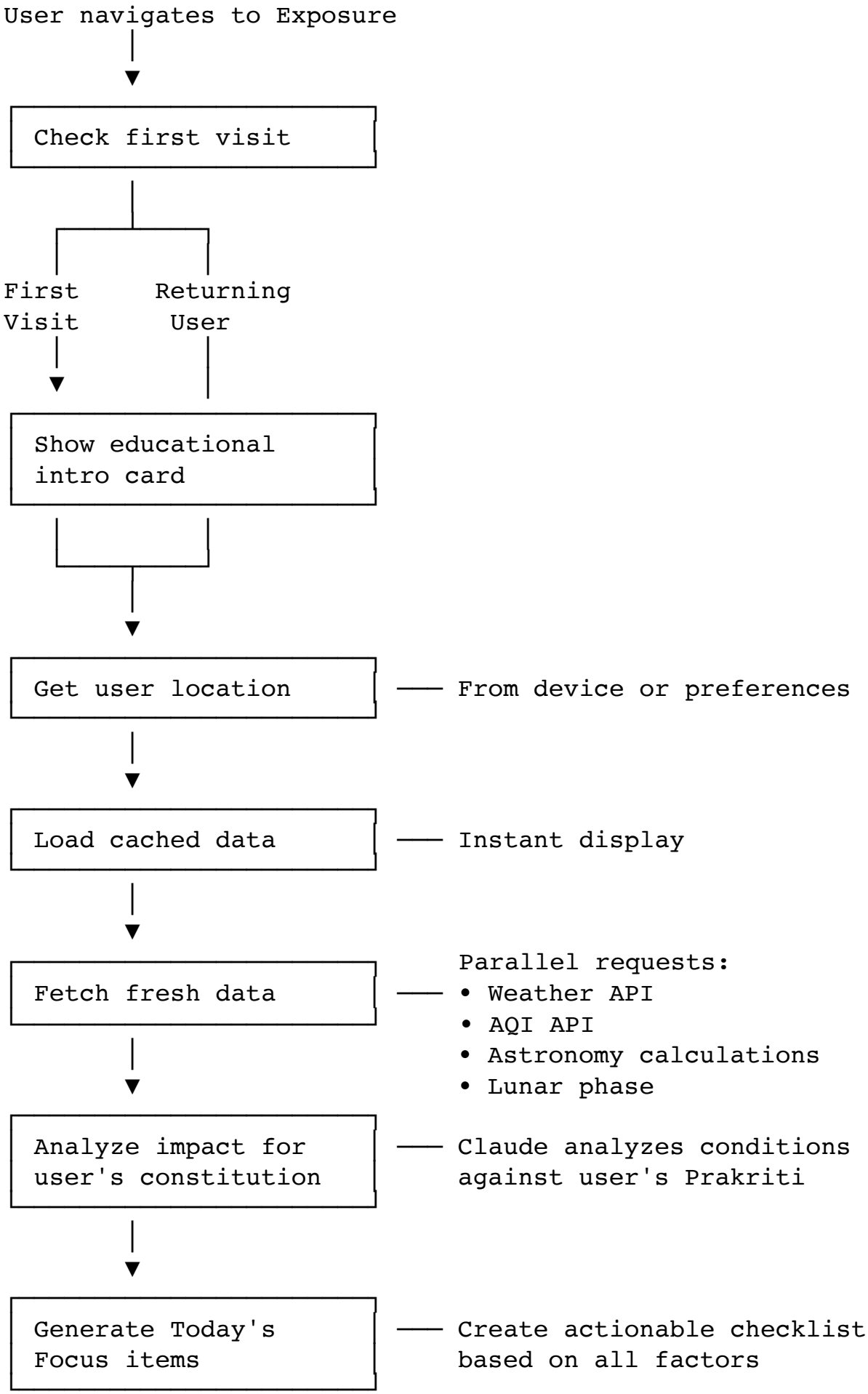
```
│   │   Exposure Agent      │   │      External Data Services
    │
    │   (Claude API)        │   │
    │
    │                       │   │  • Weather API integration
    │
    │   • Impact analysis   │   │  • AQI API integration
    │
    │   • Recommendations   │   │  • Astronomical calculations
    │
    │   • Focus generation  │   │  • Location services
    │
    │   • Protocol creation │   │  • Planetary ephemeris
    │
    └───────────────────────┘
└──────────────────────────────────────┘     │
└──────────────────────────────────────────────────────────────────────────────
──────────┘
                                             │
                                             ▼
┌──────────────────────────────────────────────────────────────────────────────
──────────┐
│                                 DATA LAYER
│
│   ┌───────────────────────┐
┌──────────────────────────────────────┐     │
│   │   Local Cache         │   │      Backend APIs
    │
    │                       │   │
    │
    │   • Weather data      │   │  • Exposure aggregation
service  │    │
│   │   • Focus completion  │   │  • Impact analysis service
    │
    │   • Preferences       │   │  • Protocol service
    │
    │   • Historical data   │   │  • Shop service
    │
    └───────────────────────┘
└──────────────────────────────────────┘     │
└──────────────────────────────────────────────────────────────────────────────
──────────┘
```

4.2 External API Integrations

```
┌──────────────────────────────────────────────────────────────────────────────
──────────┐
```

```
│                        EXTERNAL DATA SOURCES
│
│
│   ┌──────────────────┐   ┌──────────────────┐
┌──────────────────┐
│   │  Weather API     │   │   AQI API        │   │  Astronomy API
│   │
│   │  (OpenWeather/   │   │  (AirVisual/     │   │  (Internal
│   │
│   │   Weather.com)   │   │   OpenAQ)        │   │
calculations) │     │
│   │                  │   │                  │   │
│   │
│   │ • Temperature    │   │ • AQI value      │   │ • Sunrise/set
│   │
│   │ • Humidity       │   │ • PM2.5          │   │ • UV index
│   │
│   │ • Conditions     │   │ • Pollutants     │   │ • Moon phase
│   │
│   │ • Forecast       │   │ • Health recs    │   │ • Planetary
pos │     │
│   └──────────────────┘   └──────────────────┘
└──────────────────┘       │
│
│
│   ┌──────────────────┐   ┌──────────────────┐
│   │  Location        │   │  Light Sensors   │
│   │  Services        │   │  (Device)        │
│   │                  │   │                  │
│   │ • GPS coords     │   │ • Ambient lux    │
│   │ • City name      │   │ • Light type     │
│   │ • Timezone       │   │                  │
│   └──────────────────┘   └──────────────────┘
│
└──────────────────────────────────────────────────────────────────────
──────────┘
```

4.3 Data Flow: Opening Exposure Section


User navigates to Exposure
Check first visit
First Visit
Returning User
Show educational intro card
Get user location
From device or preferences
Load cached data
Instant display
Fetch fresh data
Parallel requests:
• Weather API
• AQI API
• Astronomy calculations
• Lunar phase
Analyze impact for user's constitution
Claude analyzes conditions against user's Prakriti
Generate Today's Focus items
Create actionable checklist based on all factors

```
        │
        ▼
┌──────────────────────────┐
│ Render all cards         │
```

4.4 Data Flow: Today's Focus Item Completion

```
User taps checkbox
        │
        ▼
┌──────────────────────────┐
│ Optimistic update        │ ── Immediately show checked
        │
        ▼
┌──────────────────────────┐
│ Animate checkbox         │ ── Fill animation (200ms)
│ and strikethrough        │
        │
        ▼
┌──────────────────────────┐
│ Update progress          │ ── "2/4 completed"
        │
        ▼
┌──────────────────────────┐
│ Persist locally          │ ── Save completion state
        │
        ▼
┌──────────────────────────┐
│ Sync to backend          │ ── PATCH /exposure/focus/complete
```

4.5 Data Refresh Strategy

```
Data Refresh Intervals:

┌──────────────────────────────────────────────────────────────────────────────
──────────┐
│  Data Type            │ Refresh Interval │ Trigger
│
├──────────────────────────────────────────────────────────────────────────────
──────────┤
│  Weather              │ Every 30 min     │ Timer + Pull-to-
refresh   │
```

```
│  AQI                  │ Every 60 min     │ Timer + Pull-to-
refresh   │
│  UV Index             │ Every 15 min     │ Timer + Pull-to-
refresh   │
│  Lux Level            │ Real-time        │ Device sensor (if
avail)  │
│  Sunrise/Sunset       │ Daily            │ On app open
│
│  Moon Phase           │ Daily            │ On app open
│
│  Planetary            │ Daily            │ On app open
│
│  Today's Focus        │ Daily reset      │ Midnight local
time       │
└──────────────────────────────────────────────────────────────
──────────┘
```

5. AI Integration & Prompting Strategy

5.1 The Role of Claude in Exposure Section

Claude serves several functions:

Capability	Description
Impact Analysis	Interpret conditions for specific constitution
Symptom Prediction	Predict likely symptoms based on environmental factors
Recommendation Generation	Create personalized guidance for conditions
Focus Item Generation	Build actionable checklist from all factors
Protocol Creation	Generate weather/lunar/circadian protocols
Educational Content	Explain Ayurvedic principles of environment

5.2 System Prompt for Exposure Section

```
You are the Ayurvedic Digital Twin for AyuAware, guiding the
user through
the Exposure section. Your role is to interpret environmental
conditions
through an Ayurvedic lens and provide personalized guidance.

## YOUR ROLE IN THIS SECTION
```

You are an environmental interpreter. You take weather data, light levels,
lunar phases, and (optionally) planetary positions and explain how they
specifically affect THIS user based on their constitution.

ENVIRONMENTAL PRINCIPLES

Weather affects doshas differently:

HEAT (Hot weather):
- Aggravates: Pitta (inflammation, irritability, skin issues)
- Pacifies: Vata and Kapha

COLD (Cold weather):
- Aggravates: Vata (anxiety, stiffness, dryness)
- Pacifies: Pitta

DRY (Low humidity):
- Aggravates: Vata (dry skin, constipation, restlessness)
- Pacifies: Kapha

HUMID (High humidity):
- Aggravates: Kapha (congestion, sluggishness)
- Slightly aggravates: Pitta (if combined with heat)
- Pacifies: Vata

WIND:
- Aggravates: Vata (anxiety, scattered thoughts)
- Neutral for others

LIGHT EXPOSURE PRINCIPLES

- Morning sunlight (before 9 AM): Regulates circadian rhythm
- Midday sun: Vitamin D production but can aggravate Pitta
- Evening light: Should dim for healthy sleep
- Blue light (screens): Disrupts melatonin, aggravates Pitta

LUNAR PRINCIPLES

- Waxing (New → Full): Energy builds, good for new projects
- Full Moon: Peak energy, can aggravate Pitta
- Waning (Full → New): Energy decreases, focus on completion
- New Moon: Rest, introspection, lowest energy

```
Moon governs Kapha and water element in the body.

## COMMUNICATION APPROACH

When describing impact:
- Lead with personal relevance
- Explain the mechanism briefly
- List specific symptoms to watch for
- Always provide actionable recommendations

When creating recommendations:
- Be specific and actionable
- Prioritize by importance
- Connect to constitution
- Include timing where relevant

## WHAT NOT TO DO

- Don't just report conditions — interpret them
- Don't be alarmist about weather
- Don't ignore the user's constitution
- Don't give generic advice — personalize
- Don't mix astrology with Ayurveda unless user has it
enabled

## ASTROLOGICAL CONTENT

Planetary information is OPTIONAL. Only discuss if:
1. User has planetaryEnabled = true in preferences
2. User explicitly asks about astrology

If disabled, don't mention planetary influences at all.
```

5.3 Context Building for Exposure Section

Every Claude call includes:

```
CONTEXT PACKAGE STRUCTURE:

## User Profile
- Name: {user_name}
- Location: {city, country}
- Prakriti: {prakriti_percentages}
- Current Vikriti: {vikriti_percentages}
- Active imbalances: {imbalance_list}
```

```
## Current Environmental Data
- Temperature: {temp}°C (feels like {feels_like}°C)
- Humidity: {humidity}%
- Conditions: {weather_description}
- Wind: {wind_speed} km/h from {wind_direction}
- Pressure: {pressure} hPa ({trend})

## Air Quality
- AQI: {aqi_value} ({aqi_category})
- Primary pollutant: {pollutant}
- Health guidance: {aqi_guidance}

## Light Data
- Current UV: {uv_index} ({uv_category})
- Peak UV today: {peak_uv} at {peak_time}
- Current lux: {lux_level}
- Sunrise: {sunrise_time}
- Sunset: {sunset_time}
- Current time: {current_time}

## Lunar Data
- Phase: {moon_phase}
- Day of cycle: {day}/29
- Illumination: {illumination}%
- Next significant phase: {next_phase} in {days} days

## Planetary Data (if enabled)
- Sun sign: {sun_sign}
- Major transits: {transits}
- Rahu Kala today: {rahu_kala}

## User Preferences
- Planetary info enabled: {true/false}
- Previous completions today: {completed_focus_items}
```

5.4 Prompt Templates by Scenario

Environmental Impact Analysis:

```
Analyze today's environmental conditions for this user.

{CONTEXT_PACKAGE}

Generate an impact analysis that includes:

1. **Primary Impact** (1-2 sentences)
```

```
   - Which dosha is most affected
   - Brief explanation of why

2. **Symptoms to Watch For** (4-5 bullets)
   - Specific to their constitution
   - Based on current conditions

3. **Recommendations** (5-6 bullets)
   - Actionable and specific
   - Prioritized by importance
   - Include timing where relevant

Tone: Informative but not alarmist
Personalization: Specific to their Prakriti and current
Vikriti
```

Today's Focus Generation:

```
Generate Today's Focus checklist items based on all
environmental factors.

{CONTEXT_PACKAGE}

Generate 3-5 priority action items that:
1. Address the most important environmental factors
2. Are specific and actionable
3. Include brief explanation of why
4. Are achievable in one day
5. Consider user's constitution

Format each item as:
- Title (action verb + specific task)
- Subtitle (brief explanation of why)

Prioritize items by health impact.
```

Light Exposure Recommendations:

```
Generate circadian health recommendations based on light
data.

{CONTEXT_PACKAGE}

Include recommendations for:
1. Morning light exposure
2. Midday sun considerations
3. Evening light management
4. Device/screen usage
```

Consider:
- Current UV levels
- User's constitution (Pitta = more sun sensitive)
- Time of year
- Their location

Lunar Impact Analysis:

Explain the current moon phase's impact on health.

{CONTEXT_PACKAGE}

Include:
1. What this phase means generally
2. How it affects each dosha
3. Health effects specific to their constitution
4. Practical recommendations
5. How long this phase lasts

Tone: Educational, connected to tradition
Avoid: Being overly mystical or unscientific

Environmental Protocol Generation:

Generate a protocol to balance current environmental conditions.

{CONTEXT_PACKAGE}

Create a protocol with:

1. NUTRITION (4-5 items)
   - Foods that balance the environmental impact
   - Foods to avoid
   - Beverage recommendations

2. SUPPLEMENTS (2-3 items)
   - Specific to conditions
   - Include dosage guidance

3. LIFESTYLE (4-5 items)
   - Daily practices
   - Timing recommendations
   - Cooling/warming techniques as needed

4. METADATA
   - Duration (how long to follow)

```
   - Difficulty level

Make everything specific to their constitution and current
conditions.
```

Planetary Analysis (if enabled):

```
Analyze current planetary transits from Vedic astrology
perspective.

{CONTEXT_PACKAGE}

Include:
1. Current major transits
2. How they may affect this user
3. Health-related considerations
4. Favorable times today
5. Times to avoid (Rahu Kala)

Keep grounded and practical. Focus on timing and energy
management
rather than predictions or fate.
```

5.5 Impact Alert Logic

Claude determines which dosha is most affected by analyzing conditions:

```
IMPACT DETERMINATION LOGIC:

1. Score each condition for dosha impact:

   Temperature:
   - > 30°C: Pitta +2
   - > 25°C: Pitta +1
   - < 15°C: Vata +2, Kapha +1
   - < 5°C: Vata +3, Kapha +2

   Humidity:
   - > 80%: Kapha +2, Pitta +1
   - > 60%: Kapha +1
   - < 30%: Vata +2
   - < 20%: Vata +3

   Wind:
   - > 30 km/h: Vata +2
   - > 15 km/h: Vata +1

   Weather:
```

```
   - Sunny: Pitta +1
   - Cloudy: Neutral
   - Rainy: Kapha +1
   - Stormy: Vata +2

2. Total scores and determine primary impact
3. Cross-reference with user's constitution:
   - If impacted dosha matches their dominant Prakriti →
higher alert
   - If already imbalanced (Vikriti) → highest alert

4. Generate appropriate alert level and message
```

5.6 Educational Content by Topic

Weather-Dosha Relationships:

Condition	Vata Effect	Pitta Effect	Kapha Effect
Hot	Pacifies ✓	Aggravates ⚠️	Pacifies ✓
Cold	Aggravates ⚠️	Pacifies ✓	Aggravates ⚠️
Dry	Aggravates ⚠️	Neutral	Pacifies ✓
Humid	Pacifies ✓	Aggravates ⚠️	Aggravates ⚠️
Windy	Aggravates ⚠️	Neutral	Neutral

Moon Phase Health Framework:

Phase	Energy	Best For	Avoid
New Moon	Lowest	Rest, introspection	Starting projects
Waxing	Building	New initiatives	Overexertion
Full Moon	Peak	Completion, celebration	Fasting
Waning	Decreasing	Finishing, releasing	New starts

6. Data Models & State Management

6.1 Core Data Types

Environmental Data:

```
EnvironmentalConditions:
  - temperature: number (°C)
  - feelsLike: number (°C)
  - humidity: number (%)
  - conditions: WeatherCondition
  - conditionsDescription: string
  - windSpeed: number (km/h)
  - windDirection: string
  - pressure: number (hPa)
  - pressureTrend: 'rising' | 'falling' | 'stable'
  - visibility: number (km)
  - cloudCover: number (%)
  - precipitationChance: number (%)
  - lastUpdated: Date

WeatherCondition:
  'clear' | 'partly_cloudy' | 'cloudy' | 'overcast' |
  'light_rain' | 'rain' | 'heavy_rain' | 'thunderstorm' |
  'snow' | 'fog' | 'haze' | 'windy'

AirQuality:
  - aqi: number (0-500)
  - category: 'good' | 'moderate' | 'unhealthy_sensitive' |
              'unhealthy' | 'very_unhealthy' | 'hazardous'
  - primaryPollutant: string
  - pm25: number
  - pm10: number
  - ozone: number
  - healthGuidance: string
  - lastUpdated: Date
```

Light Data:

```
LightExposure:
  - currentUV: number (0-11+)
  - uvCategory: 'low' | 'moderate' | 'high' | 'very_high' |
'extreme'
  - peakUV: number
  - peakUVTime: string (HH:mm)
  - uvProtectionNeeded: { start: string, end: string }
  - currentLux: number
  - luxCategory: string
  - sunrise: string (HH:mm)
  - sunset: string (HH:mm)
  - daylightHours: number
```

```
  - brahmaMuhurta: { start: string, end: string }
  - twilight: { start: string, end: string }

LightRecommendation:
  - category: 'morning' | 'midday' | 'evening' | 'screens'
  - recommendation: string
  - timing: string
  - priority: 'high' | 'medium' | 'low'
```

Lunar Data:

```
LunarPhase:
  - phase: MoonPhase
  - phaseName: string
  - dayOfCycle: number (1-29)
  - illumination: number (%)
  - nextNewMoon: Date
  - nextFullMoon: Date
  - doshaEffects: DoshaEffect[]
  - healthEffects: string[]
  - recommendations: string[]

MoonPhase:
  'new_moon' | 'waxing_crescent' | 'first_quarter' |
'waxing_gibbous' |
  'full_moon' | 'waning_gibbous' | 'last_quarter' |
'waning_crescent'

DoshaEffect:
  - dosha: 'vata' | 'pitta' | 'kapha'
  - effect: 'increased' | 'decreased' | 'stable'
  - description: string
```

Planetary Data (Optional):

```
PlanetaryData:
  - enabled: boolean
  - sunSign: string
  - moonSign: string
  - majorTransits: Transit[]
  - rahuKala: { start: string, end: string }
  - favorableTimes: TimeWindow[]
  - unfavorableTimes: TimeWindow[]

Transit:
  - planet: string
  - sign: string
  - description: string
```

```
  - healthEffects: string[]
  - recommendations: string[]

TimeWindow:
  - start: string (HH:mm)
  - end: string (HH:mm)
  - activity: string
  - reason: string
```

Today's Focus:

```
TodaysFocus:
  - items: FocusItem[]
  - progress: number (completed count)
  - total: number
  - generatedAt: Date
  - resetAt: Date (midnight local)

FocusItem:
  - id: string
  - title: string
  - subtitle: string
  - source: 'weather' | 'uv' | 'lunar' | 'planetary' |
'constitution'
  - priority: 'high' | 'medium' | 'low'
  - completed: boolean
  - completedAt: Date | null
```

Environmental Impact:

```
EnvironmentalImpact:
  - primaryDosha: 'vata' | 'pitta' | 'kapha'
  - severity: 'low' | 'moderate' | 'high'
  - explanation: string
  - symptoms: string[]
  - recommendations: Recommendation[]
  - alertColor: string (hex)

Recommendation:
  - action: string
  - timing: string (optional)
  - priority: 'high' | 'medium' | 'low'
  - category: 'nutrition' | 'lifestyle' | 'practice' |
'supplement'
```

6.2 State Structure

```
ExposureState:
  # Location
```

```
  - location: UserLocation
  - locationPermission: 'granted' | 'denied' | 'pending'

  # Environmental Data
  - conditions: EnvironmentalConditions | null
  - airQuality: AirQuality | null
  - isLoadingConditions: boolean
  - conditionsError: Error | null

  # Light Data
  - lightExposure: LightExposure | null
  - isLoadingLight: boolean

  # Lunar Data
  - lunarPhase: LunarPhase | null
  - isLoadingLunar: boolean

  # Planetary Data
  - planetaryData: PlanetaryData | null
  - planetaryEnabled: boolean
  - isLoadingPlanetary: boolean

  # Impact Analysis
  - impact: EnvironmentalImpact | null
  - isAnalyzingImpact: boolean

  # Today's Focus
  - todaysFocus: TodaysFocus | null
  - isLoadingFocus: boolean

  # UI State
  - isFirstVisit: boolean
  - expandedCards: string[]
  - scrollPosition: number

  # Chat
  - messages: ExposureMessage[]
  - isTyping: boolean
```

6.3 State Management Approach

Zustand Stores:

```
Stores:

1. environmentStore
```

```
   - Weather conditions
   - Air quality
   - Refresh logic
   - API integration

2. cyclesStore
   - Light exposure data
   - Lunar phase data
   - Planetary data (if enabled)
   - Astronomical calculations

3. focusStore
   - Today's Focus items
   - Completion tracking
   - Daily reset logic

4. exposureUIStore
   - First visit flag
   - Expanded cards
   - Scroll position
   - Messages
   - Preferences (planetary enabled)
```

7. Component Requirements

7.1 Environmental Components

EnvironmentalImpactCard

Displays current conditions with personal impact:

- Animated weather icon (80x80px)
- Conditions summary text
- Dosha impact alert (color-coded)
- Quick metrics row (temp, humidity, AQI)
- Expand button

States:

- Collapsed: Weather, impact alert, key metrics
- Expanded: Full details, symptoms, recommendations

WeatherIcon

- Animated icons for each condition
- Subtle movement (clouds drift, rain falls)
- Size: 80x80px
- Custom illustrations for:
  - Clear, partly cloudy, cloudy, overcast
  - Light rain, rain, heavy rain, thunderstorm

    - Snow, fog, haze, windy

DoshaImpactAlert

- Color-coded by dosha:
    - Vata: #7AB8AC (light teal)
    - Pitta: #E89B7F (coral)
    - Kapha: #A3B899 (sage green)
- Warning icon
- Brief explanation
- Appears within cards

MetricsRow

- Horizontal layout with separators
- Icon + value + unit for each metric
- Tappable for more detail

7.2 Light Exposure Components

LightExposureCard

- Sun arc visualization
- Sunrise/sunset times
- Current UV and lux
- Insight text
- Expandable detail

SunArcVisualization

- Curved arc showing sun's path across sky
- Current position marked with sun icon
- Sunrise and sunset labeled at ends
- Color gradient (gold to orange)
- Size: Full width, 100px height
- Animated sun position updates

UVCurveChart

- Line chart showing UV throughout day
- Current time marked
- Peak time highlighted
- Protection needed zone shaded
- Size: Full width, 80px height

7.3 Today's Focus Components

TodaysFocusCard

- Priority action checklist
- Interactive checkboxes
- Progress indicator
- "Mark All Complete" button

ChecklistItem

- Height: 60px
- Checkbox: 24x24px
- Title: 15px, weight 600

- Subtitle: 13px, weight 400, color #5A6B66
- Border bottom between items

Interactive States:

- Unchecked: Empty checkbox, normal text
- Checked: Filled checkbox (#4A9B8E), strikethrough text
- Tap animation: Checkbox fills (200ms)

ProgressIndicator

- Text: "2/4 completed"
- Optional: Progress bar below

7.4 Lunar Components

LunarCard

- Moon phase visualization
- Phase name
- Impact summary
- Expandable detail

MoonPhaseVisualization

- Accurate visual of current phase
- Size: 100x100px
- Subtle glow effect
- Real-time phase calculation
- Illumination percentage shown

LunarCycleStrip

- 8 phase icons in a row
- Current phase highlighted
- Shows full cycle at a glance
- Tappable for dates

7.5 Planetary Components (Optional)

PlanetaryCard

- Astrological symbol/chart
- Current transits
- Impact summary
- "Disable" option
- Expandable detail

TransitBlock

- Planet symbol + sign
- Effects description
- Recommendations

TimingGrid

- Favorable times
- Unfavorable times (Rahu Kala)
- Time blocks with activities

7.6 Protocol Components

EnvironmentalProtocolCard

- Same structure as Health protocols
- Sections: Nutrition, Supplements, Lifestyle
- Context-specific to weather/lunar/planetary
- Save and Start buttons

7.7 Animation Requirements

Animation	Duration	Easing	Trigger
Weather icon	Continuous	linear	On display
Card expand	400ms	ease-out	Tap expand
Content fade-in	300ms	ease-in	After expand
Checkbox fill	200ms	ease-out	Tap checkbox
Strikethrough	200ms	ease-out	After checkbox
Sun position update	500ms	ease-in-out	Data refresh
Moon glow	3000ms	ease-in-out	Continuous (subtle)

8. API Requirements

8.1 REST Endpoints

Get All Exposure Data:

```
GET /api/v1/exposure/current

Response:
{
  "conditions": {
    "temperature": 28,
    "feelsLike": 32,
    "humidity": 81,
    "conditions": "partly_cloudy",
    "conditionsDescription": "Partly cloudy",
    "windSpeed": 8,
    "windDirection": "SW",
    "pressure": 1013,
    "lastUpdated": "2025-01-31T10:30:00Z"
  },
  "airQuality": {
    "aqi": 44,
    "category": "good",
    "primaryPollutant": "PM2.5",
```

```
    "healthGuidance": "Air quality is satisfactory."
  },
  "light": {
    "currentUV": 2,
    "uvCategory": "low",
    "peakUV": 7,
    "peakUVTime": "13:00",
    "currentLux": 1000,
    "sunrise": "06:41",
    "sunset": "19:42",
    "daylightHours": 13.02
  },
  "lunar": {
    "phase": "waning_crescent",
    "phaseName": "Waning Crescent",
    "dayOfCycle": 26,
    "illumination": 15,
    "nextNewMoon": "2025-02-03"
  },
  "impact": {
    "primaryDosha": "pitta",
    "severity": "moderate",
    "explanation": "High heat and humidity may aggravate
Pitta...",
    "symptoms": [...],
    "recommendations": [...]
  },
  "location": {
    "city": "Dharamsala",
    "country": "India",
    "timezone": "Asia/Kolkata"
  }
}
```


Get Weather Forecast:

```
GET /api/v1/exposure/forecast?days=7

Response:
```


```
{
  "forecast": [
    {
      "date": "2025-02-01",
      "high": 29,
      "low": 18,
      "conditions": "partly_cloudy",
```

```
      "precipitationChance": 20,
      "uvIndex": 7,
      "impact": { ... }
    },
    ...
  ]
}
```

Get Today's Focus:

```
GET /api/v1/exposure/focus

Response:
{
  "items": [
    {
      "id": "focus_001",
      "title": "Take Vitamin D After Lunch",
      "subtitle": "UV is low, supplement recommended",
      "source": "uv",
      "priority": "high",
      "completed": false
    },
    ...
  ],
  "progress": 0,
  "total": 4,
  "generatedAt": "2025-01-31T00:00:00Z"
}
```

Update Focus Item:

```
PATCH /api/v1/exposure/focus/complete

Request:
{
  "itemId": "focus_001",
  "completed": true
}

Response:
{
  "success": true,
  "progress": 1,
  "total": 4
}
```

Get Lunar Details:

```
GET /api/v1/exposure/lunar

Response:
{
  "phase": { ... },
  "doshaEffects": [
    { "dosha": "vata", "effect": "stable", "description":
"..." },
    { "dosha": "pitta", "effect": "decreased", "description":
"..." },
    { "dosha": "kapha", "effect": "increased", "description":
"..." }
  ],
  "healthEffects": [...],
  "recommendations": [...],
  "ayurvedicPerspective": "...",
  "fullCycle": [
    { "phase": "new_moon", "date": "2025-02-03" },
    ...
  ]
}
```

Get Planetary Data (if enabled):

```
GET /api/v1/exposure/planetary

Response:
{
  "enabled": true,
  "sunSign": "Gemini",
  "moonSign": "Pisces",
  "majorTransits": [
    {
      "planet": "Sun",
      "sign": "Gemini",
      "description": "Mercury rules Gemini...",
      "healthEffects": [...],
      "recommendations": [...]
    }
  ],
  "rahuKala": { "start": "15:00", "end": "16:30" },
  "favorableTimes": [...],
  "unfavorableTimes": [...]
}
```

Update Exposure Preferences:

```
PATCH /api/v1/exposure/preferences
```

```
Request:
{
  "planetaryEnabled": false
}

Response:
{
  "success": true,
  "preferences": {
    "planetaryEnabled": false
  }
}
```

Create Environmental Protocol:

```
POST /api/v1/exposure/protocols/create

Request:
{
  "type": "weather",
  "conditions": {
    "temperature": 28,
    "humidity": 81,
    "impactedDosha": "pitta"
  }
}

Response:
{
  "protocol": {
    "id": "proto_env_xxx",
    "name": "Pitta-Balancing Weather Protocol",
    "nutrition": { ... },
    "supplements": [ ... ],
    "lifestyle": [ ... ],
    "duration": "While hot weather lasts",
    "difficulty": "easy"
  },
  "recommendedProducts": [...]
}
```

8.2 External API Integrations

Weather APIs (choose one):

- OpenWeatherMap API
- Weather.com API

- AccuWeather API

Air Quality APIs (choose one):

- AirVisual API
- OpenAQ API
- World Air Quality Index API

Astronomical Data:

- Internal calculations for sunrise/sunset
- Internal calculations for moon phase
- Optional: Swiss Ephemeris for planetary positions

8.3 WebSocket Events (Optional)

```
WebSocket Events:

conditions.updated
{
  "temperature": 29,
  "humidity": 80,
  "conditions": "partly_cloudy",
  "updatedAt": "2025-01-31T11:00:00Z"
}

focus.completed
{
  "itemId": "focus_001",
  "completed": true,
  "progress": 2,
  "total": 4
}

focus.reset
{
  "items": [...],
  "generatedAt": "2025-02-01T00:00:00Z"
}
```

9. Success Metrics

9.1 Engagement Metrics

Metric	Target	Measurement
Section visits per week	> 4	Users visiting Exposure weekly
Card expansion rate	> 50%	Expanded cards / shown cards

Time in section	> 2 minutes	Average session duration
Return rate	> 50%	Users who return within 7 days
Pull-to-refresh usage	Track	Manual refresh frequency

9.2 Focus Checklist Metrics

Metric	Target	Measurement
Focus item completion rate	> 40%	Items completed / items generated
Full checklist completion	> 15%	Users who complete all items
Daily engagement	> 30%	Users who check at least 1 item
Completion time	Track	When items are typically checked

9.3 Action Metrics

Metric	Target	Measurement
Protocol creation rate	> 15%	Users who create environmental protocol
Shop click-through	> 10%	Users who tap shop links
Lunar card engagement	> 40%	Users who expand lunar card
Planetary opt-out rate	Track	Users who disable astrology

9.4 Data Quality Metrics

Metric	Target	Measurement
Location accuracy	> 95%	Correct city resolution
Weather data freshness	< 30 min	Time since last update
API reliability	> 99%	Successful API calls
Impact relevance	Survey	User rating of recommendations

9.5 Technical Metrics

Metric	Target	Measurement
Section load time	< 2s	Time to first meaningful content
Weather refresh time	< 1s	API call duration

Focus item update	< 500ms	Time to reflect checked state
Error rate	< 1%	Failed loads or updates

10. Implementation Considerations

10.1 Location Services

Location Acquisition:

1. Request location permission on first use
2. Use GPS for precise location
3. Fall back to IP-based geolocation
4. Allow manual location override in settings
5. Cache location for subsequent sessions

Privacy Considerations:

- Explain why location is needed
- Allow app use without location (reduced functionality)
- Don't store location history beyond current session
- Provide location privacy controls

10.2 API Rate Limiting

Strategy:

- Cache weather data for 30 minutes
- Cache AQI data for 60 minutes
- Cache lunar/planetary data for 24 hours
- Implement request queuing
- Show cached data while refreshing

Fallback:

- If weather API fails, show last known data
- Display "Last updated X hours ago" warning
- Provide manual refresh option

10.3 Offline Handling

When Offline:

- Show cached weather data with timestamp
- Show cached focus items (completion still works)
- Lunar and planetary data fully available (calculated locally)
- Queue focus completions for sync
- Disable protocol creation

When Back Online:

- Sync queued completions
- Refresh weather data
- Update impact analysis

10.4 Performance Optimization

Data Loading:

- Load cached data immediately
- Fetch fresh data in background
- Update UI incrementally as data arrives
- Lazy-load expanded card content

Animations:

- Use CSS transforms for weather icons
- Throttle sun arc updates
- Reduce animations on low-end devices
- Respect reduced motion preference

10.5 Accessibility Requirements

Screen Reader:

- Weather conditions fully announced
- Impact alert announced with severity
- Focus items announced with completion state
- Moon phase announced with description

Motor Accessibility:

- All tap targets ≥ 44x44px
- Checkboxes have generous tap area
- Swipe alternatives for refresh
- No time-sensitive interactions

Visual Accessibility:

- WCAG AA contrast compliance
- Don't rely on color alone for impact
- Support Dynamic Type
- Weather icons have text labels

10.6 Analytics Events

Event	Properties	Purpose
`exposure_opened`	source, is_first_visit	Track entry
`conditions_viewed`	temp, humidity, impact	Environmental engagement
`conditions_expanded`	—	Detail engagement
`focus_item_completed`	item_id, source	Action tracking
`focus_all_completed`	item_count	Full completion
`lunar_expanded`	phase	Cycle interest
`planetary_expanded`	—	Astrology interest
`planetary_disabled`	—	Opt-out tracking
`protocol_created`	type (weather/lunar)	Action taken
`shop_clicked`	product_category	Commerce interest

`location_granted`	—	Permission tracking
`location_denied`	—	Permission tracking

10.6 Testing Priorities

Unit Tests:

- Moon phase calculation
- Dosha impact determination
- UV category mapping
- Focus item generation logic

Integration Tests:

- Weather API integration
- AQI API integration
- Focus completion sync
- Location service integration

E2E Tests:

- First-time user flow with location permission
- Weather → Impact → Protocol creation
- Focus checklist completion
- Lunar card exploration

Manual QA:

- Weather icon accuracy for all conditions
- Impact relevance for different constitutions
- Lunar phase accuracy
- Recommendation quality
- Protocol content relevance

Appendix A: Weather-Dosha Reference

Temperature Impact

Temperature	Vata	Pitta	Kapha
> 35°C	Pacifies ✓	Aggravates ⚠️⚠️	Pacifies ✓
25-35°C	Pacifies ✓	Aggravates ⚠️	Pacifies ✓
15-25°C	Neutral	Neutral	Neutral
5-15°C	Aggravates ⚠️	Pacifies ✓	Aggravates ⚠️
< 5°C	Aggravates ⚠️⚠️	Pacifies ✓	Aggravates ⚠️⚠️

Humidity Impact

Humidity	Vata	Pitta	Kapha
> 80%	Pacifies ✓	Aggravates ⚠️	Aggravates ⚠️⚠️
50-80%	Pacifies ✓	Neutral	Aggravates ⚠️
30-50%	Neutral	Neutral	Neutral
< 30%	Aggravates ⚠️⚠️	Neutral	Pacifies ✓

Combined Conditions

Condition	Most Affected	Recommendation Theme
Hot + Humid	Pitta, Kapha	Cooling, light foods
Hot + Dry	Pitta	Cooling, hydrating
Cold + Dry	Vata	Warming, moistening
Cold + Humid	Kapha, Vata	Warming, stimulating
Windy	Vata	Grounding, calming

Appendix B: Moon Phase Reference

Eight Phases

Phase	Day	Illumination	Energy	Best For
New Moon	0	0%	Lowest	Rest, intention setting
Waxing Crescent	1-7	1-49%	Building	Starting projects
First Quarter	7	50%	Growing	Taking action
Waxing Gibbous	7-14	51-99%	Peak building	Refinement
Full Moon	14	100%	Highest	Celebration, completion
Waning Gibbous	14-21	99-51%	Decreasing	Sharing, gratitude
Last Quarter	21	50%	Lower	Releasing
Waning Crescent	21-29	49-1%	Lowest	Rest, reflection

Dosha Effects by Phase

Phase	Vata	Pitta	Kapha
Waxing	Slightly ↑	Slightly ↑	↑↑
Full	↑	↑↑	↑↑
Waning	↓	↓	↓
New	Stable	Stable	Stable

Appendix C: AQI Reference

AQI Categories

AQI	Category	Health Implications	Guidance
0-50	Good	None	No restrictions
51-100	Moderate	Sensitive groups may react	Sensitive individuals be
101-150	Unhealthy for Sensitive	Respiratory symptoms in sensitive	Reduce outdoor exertion
151-200	Unhealthy	Everyone may have symptoms	Avoid prolonged outdoor activity
201-300	Very Unhealthy	Serious health effects	Minimize outdoor activity
301-500	Hazardous	Emergency conditions	Stay indoors

Ayurvedic Considerations

- Poor air quality aggravates **Vata** (respiratory system)
- Pollution creates **Ama** (toxins) in body
- Supports need for **Prana** enhancement indoors
- Recommends air-purifying practices (neti, pranayama)

Appendix D: UV Index Reference

UV Categories

UV Index	Category	Time to Burn	Protection Needed
0-2	Low	> 60 min	Minimal
3-5	Moderate	45 min	Some
6-7	High	30 min	Recommended
8-10	Very High	15 min	Essential
11	Extreme	< 10 min	Mandatory

Constitution-Specific Guidance

Constitution	Sun Sensitivity	Vitamin D Need	Recommended Exposure
Vata	Moderate	Higher	20-30 min morning
Pitta	High	Lower	10-15 min morning only
Kapha	Low	Moderate	30+ min, midday OK

End of Document